

CSE332 - Industry Ethics & Legal Issues

Complete Lecture Notes - Units 1 to 6

Unit 1: Ethics

CSE332 Lecture 0_Without Edu-Revolution.pptx (26 slides)

Slide 1: CSE332 | INDUSTRY ETHICS AND LEGAL ISSUES | Lecture #0 | The kick start session

Slide 2: Program Outcomes | PO10 | Communication::Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. | PO11 | Project management and finance::Demonstrate knowledge and understanding of the engineering, management principles and apply the same to one's own work, as a member or a leader in a team, manage projects efficiently in respective disciplines and multidisciplinary environments after consideration of economic and financial factors. | PO12 | Life-long learning::Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Slide 3: PSO1: Apply acquired skills in software engineering, networking, security, databases, intelligent systems, cloud computing and operating systems to adapt and deploy innovative software solutions for diverse applications. | PSO2: Apply diverse IT skills to design, develop, and evaluate innovative solutions for business environments, considering risks, and utilizing interdisciplinary knowledge for efficient real-time projects benefiting society. | Program Specific Outcomes

Slide 4: Skill Attainment | Logical Thinking | Technical Competence | Publication | Patent/ Copyright | Interpersonal Skills

Slide 5: Unit | 1 | : ETHICS | Definition of ethics, Importance of integrity, Ethics in business world, improving corporate ethics, Creating an ethical work environment, Ethical decision making, Ethics in information technology, Ethical behavior of IT professional, Common ethical issues for IT users, Supporting the ethical practices of IT users | Unit 2: INTELLECTUAL PROPERTIES | Different kinds of marks (brand names, logos, signatures, symbols, well known marks, certification marks and service marks), Patents, Importance of Patent Information in Business Development, Concept of Intellectual Property, Copyrights | , | Trademarks | Unit 3: GOVERNMENT FUNDING AND SCHEMES FOR STARTUPS | what are Startups, Startup India benefits, Resources, bank loan for start up business, Start-up India, 10000 startups -A NASSCOM Initiative, Export promotion schemes: Software Technology Parks (STPs), Special Economic Zones (SEZ) Scheme, Laws for Startups | Course Contents

Slide 6: Unit | 4 | : STARTUP IN IT | Planning of startup business in IT sector-Executive summary, General company Description, Products and services, Marketing plan, Operational plan, Management and organization, Personal Financial Statement, Startup Expenses and Capitalization, Financial Plan, Appendices, Refining the Plan, Examples of Successful Start-ups | Unit 5: COMPANIES | Introduction to IT and ITES industry (Product based, Services based), Introduction to NASSCOM, STPI, Overview on latest IT projects with global impact, Case study of an IT industry (Product based and Services based), Recent technology advancement, Current affairs related with the IT industry, Diversity in the Workforce | Unit 6: LEGAL, ETHICAL AND PROFESSIONAL ISSUES IN INFORMATION SECURITY | Introduction, Law and Ethics in Information Security, Organizational Liability and the Need for Counsel, Policy Versus Law, Cyber Crime, Cyber-crime on the rise, Cyber law of India, Need for cyber law in India | Course Contents

Slide 7: Course Assessment Model | Marks Breakup | Attendance 5 | CA 25 | MTE | 20 | ETE | 50 | Total | 100

Slide 8: AT1: | Case Based Assignment | (IPR) Lecture | 06 | (Before | MTE |) | AT2: | Test2(MCQ based) Lecture | 18 | (After | MTE) | Details of Academic Tasks | Note: No MOOC Certification is Applicable

Slide 9: AT1 –IPR Rubrics | Identification of Problem Statement and formulation of Objectives- 15 Marks | Analysis of Existing Case Studies (Study of Literature) - 15 Marks | Originality and Innovation- 20 Marks | Use of Relevant Terminology (Problem Solution) - 15 Marks | Clarity and Organization (Documentation)- 20 Marks | Presentation Skills- 15 Marks | These marks will be prorated with the marks obtained by the student out of 30 marks.

Slide 10: Text Book/ Reference Book | Text | Book | ETHICS 101 | Author: | JOHN .C MAXWELL | Publisher: | Hachette India

Slide 11: Introduction to Intellectual Property | - | <https://www.coursera.org/learn/introduction-intellectual-property> | . | Startup Entrepreneurship Specialization | <https://www.django-project.com/> | <https://www.coursera.org/specializations/startup-entrepreneurship> | .com | /. | Online Education Resources

Slide 12: Course Details | Course Code | :- | CSE332 | LTP:- | 2 0 | 0 | [2 | lectures/week |] | Credits | :- | 2

Slide 13: Unit-1: Ethics

Slide 14: Unit-2: Intellectual Properties

Slide 15: Unit-3: Government Funding and Schemes for Startup

Slide 16: Unit-4: Startup in IT

Slide 17: Unit-5: Companies

Slide 18: Unit-6: Legal, Ethical and Professional Issues in Information Security

Slide 19: Next Class | : | Ethics

Slide 20: Vision | To be a globally recognised school through excellence in teaching, learning and research for creating Computer Science professionals, leaders and entrepreneurs of future contributing to society and industry for sustainable growth.

Slide 21: Mission | To build computational skills through hands-on and practice-based learning with measurable outcomes. | To establish a strong connect with industry for in-demand technology driven curriculum. | To build the infrastructure for meaningful research around societal problems. | To nurture future leaders through research-infused education and lifelong learning. | To create smart and ethical professionals and entrepreneurs who are recognized globally

Slide 22: Revised Bloom's Taxonomy | Bloom's Taxonomy is a framework for categorizing educational goals, It is widely used by educators to create learning objectives, assessments, and activities that encourage higher-order thinking. The taxonomy is typically represented as a hierarchical model with six levels: | Remembering | : Recalling facts and basic concepts. | Example: Define, list, memorize, repeat, state. | Understanding | : Explaining ideas or concepts. | Example: Describe, explain, interpret, summarize, classify. | Applying | : Using information in new situations. | Example: Use, execute, implement, demonstrate, solve. | Analyzing | : Breaking information into parts to explore understandings and relationships. | Example: Compare, contrast, examine, test, differentiate. | Evaluating | : Justifying a decision or course of action. | Example: Critique, judge, argue, support, validate. | Creating | : Producing new or original work. | Example: Design, construct, plan, produce, invent.

Slide 23: Course outcome | Through this course students should be able to: | CO1 | :: | Demonstrate Ethics across various professional domains | CO2 | :: | Apply the understanding of intellectual property concepts in practical contexts | CO3 | :: | Analyze the benefits available under startup India initiative

Slide 24: Course outcome | Through this course students should be able to: | CO4 | :: | Explain comprehensive business plan for an IT startup | CO5 | :: | Classify latest trends to contribute effectively to dynamic landscape of IT industry | CO6 | :: | : Examine various cyber laws to address emergency digital threats

Slide 25: Program Outcomes | PO1 | Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems. | PO2 | Problem analysis::Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences. | PO3 | Design/development of solutions::Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations. | PO4 | Conduct investigations of complex problems::Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Slide 26: Program Outcomes | PO5 | Modern tool usage::Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations. | PO6 | The engineer and society::Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice. | PO7 | Environment and sustainability::Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. | PO8 | Ethics::Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice. | PO9 | Individual and team work::Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

CSE332 Lecture 3.pptx (15 slides)

Slide 1: CSE332 | INDUSTRY ETHICS AND LEGAL ISSUES

Slide 2: Poor problem statement | Good | problem statement

Slide 3: It is ideal to enlist the help of others, including stakeholders, to identify several alternative solutions to the problem. | Brainstorming with others will increase your chances of identifying a broad range of alternatives and determining the best solution. | Try not to be critical of ideas, as any negative criticism will tend to shut down the discussion, and the flow of ideas will dry up. | Simply write down the ideas as they are suggested. | IDENTIFY ALTERNATIVES

Slide 4: EVALUATE AND CHOOSE AN ALTERNATIVE

Slide 5: IMPLEMENT THE DECISION | Once an alternative is selected, it should be implemented in an | efficient, effective, and timely manner. | Why are we doing this? | What is wrong with the current way we do things? | What are the benefits of the new way for you?

Slide 6: EVALUATE THE RESULTS | Were the success criteria fully met? | Were there any unintended consequences?

Slide 7: Next Topics: | Ethics in Information Technology | Ethical behavior of IT | professional | Common | ethical issues for IT | users | Supporting | the ethical practices of IT users

Slide 8: - | Creating an ethical work environment | - Ethical decision making

Slide 9: Creating an Ethical Work Environment

Slide 10: Employees | must have a knowledgeable resource with whom they can discuss | perceived unethical | practices | Creating | an Ethical Work Environment

Slide 11: Manager's checklist for establishing an ethical work environment | Creating an Ethical Work Environment

Slide 12: DECISION

Slide 13: DECISION MAKING

Slide 14: INCLUDING ETHICAL CONSIDERATIONS IN DECISION MAKING

Slide 15: A problem statement is a clear, concise description of the issue that needs to be addressed. | It Answers- | What do people observe that causes them to think there is a problem? | Who is directly affected by the problem? | Is anyone else affected? | How often does the problem occur? | What is the impact of the problem? | How serious is the problem? | DEVELOP A PROBLEM STATEMENT

CSE332 Lecture 4.pptx (12 slides)

Slide 1: CSE332 | Industrial | Ethics | and | Legal | Issues | Ethical | Behavior | of | IT | Professional

Slide 2: Supporting | the | Ethical | Practices Of | IT | Users | Education | And | Awareness | Clear | Policies | and | Standards | Access | Control | and | Security | Ethical | Use | Of | AI | And | Automation | Collaboration | and | community Involvement

Slide 3: Supporting | the | Ethical | Practices

Slide 4: Women | Empowerment | Ethical | Aspects | of | Engineering | Practices | NEXT | TOPIC:

Slide 5: ETH ICS IN INFORMATION TECHNOLOGY | Here are some examples that raise public concern about the ethical use of information technology: | Ethical concerns in technology's workplace use. | Copyright violation harms music and films. | Spam emails enable low-cost marketing. | Hacking, plagiarism, and tracking persist.

Slide 6: Profession | A | profession | is | a | calling | that | requires | specialized | knowledge | and | often | long | and | intensive | academic preparation. | Professional | ?

Slide 7: Professional | Employee | One | who | is | engaged | in | the | performance | of | work | . | Specialized | intellectual | instruction. | Exercise | of | discretion | and | judgement.

Slide 8: Professionals | such | as | doctors, | lawyers, | and | accountants | require | advanced | training | and | experience; | they | must | exercise | Carefulness | and | judgment | in | the | course | of | their | work.

Slide 9: Are | IT | Workers Professionals? | Specialized | Knowledge | Ethical | Behaviour | Accountability | Commitment | to | Continuous Learning

Slide 10: Common | Ethical | Issues | For | IT | Users | Privacy | Concern | Cybersecurity | Risks | Misuse | of | IT | Resource | Surveillance | and | Monitoring

Slide 11: Business | Software | Alliance | (BSA) | The | Business | Software | Alliance | (BSA) | is | represents | software | a | trade | group | that | the | world's | largest | and | hardware | manufacturers. | Its | mission | unauthorized | produced | by | is | to | stop | the | copying | of | its | software | members.

Slide 12: More | than | 100 | BSA | lawyers | and | investigators | prosecute | thousands | of | cases | of | software | piracy | each | year. | BSA | investigations | are | usually | triggered | by | calls | to | the | BSA | hotline, | reports | sent | to | the | BSA | Web | site | (www.nopiracy.org), | and | referrals | from | member | companies.

Unit 2: Intellectual Property Rights

CSE332 IPR.pptx (12 slides)

Slide 1: CSE332 | INDUSTRY ETHICS AND LEGAL ISSUES

Slide 2: Design | It protects | the visual appearance of a product or part of it. | Design | rights protect the unique look of a product, including its shape, color, pattern, texture, and materials | .

Slide 3: Plant | varieties | "PVRs | protect breeders from unauthorized reproduction, production, marketing, selling, importing, and exporting of their plant varieties | . | "

Slide 4: Next Topic Copy Right Infringement

Slide 5: Intellectual | Property Rights (IPR) are legal | rights. | It protect | and encourage the development and distribution of new products and services. | IPRs | are granted to people over their | creations. | S | uch | as inventions, designs, trademarks, and creative | content. | Intellectual | property rights

Slide 6:

Slide 7: "A | patent is a legal document that gives the inventor exclusive rights to their invention for a set period of time | ." | It | protects the invention from being made, used, or sold without the inventor's permission.

Slide 8: Copyright | Copyright | is the right to copy. | Only | the creator of the work can authorize anyone to reproduce the work. | Copyright law protects creators of original material from unauthorized | use

Slide 9: Things take place under copyrights | Novels | Art | Poetry | Musical lyrics and compositions | Films | Website | content

Slide 10: Trademark | A trademark is a sign capable of distinguishing the goods or | services. | Trademarks | are protected by intellectual property rights | . | T | rademark | protection can be obtained through | registration. | It will | confer an exclusive right to the use of the registered trademark.

Slide 11: Trade-Secret | A trade secret is | a confidential piece of | information. | Secrecy | : The information is not generally known or accessible to competitors | Commercial value | : The information gives the business an economic advantage | Reasonable efforts | : The business takes steps to keep the information | secret

Slide 12: Geographical Indication | A geographical indication (GI) is a name or sign that identifies a product's geographical | origin. | GIs | are used to certify that products are made in a specific place using traditional | methods. | GIs help consumers identify quality products. | GIs help producers market their products. | GIs prevent other people from using popular product names.

CSE332 Lecture 5.pptx (12 slides)

Slide 1: CSE332 | INDUSTRY ETHICS AND LEGAL ISSUES

Slide 2: Design | It protects | the visual appearance of a product or part of it. | Design | rights protect the unique look of a product, including its shape, color, pattern, texture, and materials | .

Slide 3: Plant | varieties | "PVRs | protect breeders from unauthorized reproduction, production, marketing, selling, importing, and exporting of their plant varieties | . | "

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Slide 6:

Slide 7: "A | patent is a legal document that gives the inventor exclusive rights to their invention for a set period of time | ." | It | protects the invention from being made, used, or sold without the inventor's permission.

Slide 8: Copyright | Copyright | is the right to copy. | Only | the creator of the work can authorize anyone to reproduce the work. | Copyright law protects creators of original material from unauthorized | use

Slide 9: Things take place under copyrights | Novels | Art | Poetry | Musical lyrics and compositions | Films | Website | content

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CSE332 Lecture 6.pptx (16 slides)

Slide 1: Types of Trademarks

Slide 2: Certification | marks | Define "standard" | of | goods | and | services. | Examples of common certification marks include: | UL (Underwriters Laboratories | Fair Trade | Energy Star | . | Organic Certification

Slide 3: Shape | Marks | Type of trademark that protects the three-dimensional shape | Configuration of a product or its packaging | Granted when the shape of the product itself is distinctive and capable of functioning as a source identifier for consumers | Non-traditional trademarks because they go beyond the standard text or logo-based trademarks.

Slide 4: Pattern Mark | Trademark that protect a specific pattern or design | Pattern or design used consistently to identify the source of goods or services | Examples of Pattern Marks | Gucci GG Pattern | Louis Vuitton's Monogram Pattern

Slide 5:

Slide 6: Sound | Mark | Trademark that protect a specific sound or combination of sounds | Sounds used to identify and distinguish the goods or services of a particular brand | Segistered | when the sound is distinctive enough to be recognized by consumers

Slide 7: Loss | of | Rights | Improper | use | Genericide | Non- | use | Improper | assignment | or | licensing | Failure | to | police | Failure | to | renew | Cancellation

Slide 8: Benefits | of | Registration | of | a | trademark | Registration | is | prima | facie | evidence | of | proprietorship | of | the | trademark | It | gives | the | exclusive | right | to | use | the | trademark | in | respect | of | goods/services; | and | to | take | legal | action | in | case | of | infringement | The | registered | proprietor | may | assign | or | license | the | trademark | as | any | other | property | The | registered | proprietor | can | enjoy | the | Goodwill | associated | with | registered | trademark | forever | , | if | the | mark | is | renewed | from | time | to | time

Slide 9:

Slide 10: Product | Mark | Similar | to | trademark | Identify | the | products | or | goods | instead | of | services | Product | is | the | unique | selling | feature | of | the | company | Initially, | TM, | which | states | that | the | mark | is | not | yet | registered. | Once | they | are | registered, | they | have | to | use | ®. | Examples | : | Pepsi®, | Maggi®, | PHILIPS® | etc.

Slide 11: Examples of Product Marks: | Apple's Logo | Nike's Swoosh | Coca-Cola

Slide 12: Service | Mark | I | identify | and | distinguish | the | services | rather | than | the | products | S | ymbol | is | SM | and | not | TM. | Features of a Service Mark | Service Identification | Source Indicator | Non-Tangible Association

Slide 13: Examples of Service Marks: | FedEx: | McDonald's Golden Arches | Netflix | American Airlines Logo

Slide 14:

Slide 15: Collective | Mark | A | collective | mark | is | owned | by | a | collective | I | identify | their | goods and | services | D | istinguish | their | goods | and | services | from | those | of | non- | members | I | ndicate | membership | in | the | group.

Slide 16: Examples

CSE332 Lecture 7.pptx (16 slides)

Slide 1: CSE332 | INDUSTRY ETHICS AND LEGAL ISSUES

Slide 2: Meaning | w.r.t. | to | Patents: | Any | description | of | the invention | which | destroys | the | element | of | novelty | of | the | invention

Slide 3: Who | can | file | a | patent? | Section | 6: | Persons | entitled | to | apply | for | patents | True | and | first | inventor | of | the | invention | Assignee | of | true | and | first | inventor | Legal | Representative | of | a | deceased | person | who | immediately | before | his | death | was | entitled | to | make | such | application | Application | submitted | either | alone | or | jointly | with | other | person.

Slide 4: Types | of | Applications | Ordinary | Applications | Convention | Application | Patent | of | Addition | Application | Divisional | Applications | PCT | International | Phase | Application | PCT | National | Phase | Application

Slide 5: 07-10- | 2018 | Documents | required | for | filing | of | A | Patent | Application | Application for | Grant | of | Patent | in | Form 1 | in | duplicate | Complete/Provisional | specification | in | Form | 2 | in | duplicate | Statement | and | Undertaking | in | Form | 3. | Power | of | Attorney | in | Form 26 | (in | original) | ; | (if | filed | through | attorney) | Declaration | of | Inventor-ship | in | Form | 5 | Requisite | Statutory | fees | (copy | of | the | Priority | cheque | / | DD). | Covering | letter- | indicating | the | list | of | documents

Slide 6: Contents | of | Specification | Title | of | the | invention | Field | of | the | invention | Background | of | the | invention | (PRIOR | ART) | Object | of | the | invention | Summary | of | the | invention | Brief | description | of | drawings, | if | any | Detailed | description | of | the | invention | Examples | Claims- | not | required | in | provisional | Abstract- | not | required | in | provisional

Slide 7: Provisional | application | Get | early | priority | Not | expensive | Quick | Title/Description | No | claims | Cannot | be | filed | in | case | of | conventional/PCT/Divisional | application | Filing | provisional | application | is | an | optional | but | an | advisable | step

Slide 8: C | omplete | specifications | Where | an | application | for | a | patent | is | accompanied | by | a | provisional | specification, | a | complete | specification | shall | be | filed | within | twelve | months | from | the | date | of | filing | of | the | application., | E.g. | Example | Provisional | application: | 01-12- | 2008 | Complete | specification: | 01- | 12- | 2009

Slide 9: Patents

Slide 10: Patent | : A statutory grant by the government for new inventions. | Monopoly Rights | : Grants exclusive rights to the inventor for a defined period. | Disclosure Requirement | : The inventor must fully disclose the invention. | Legal Protection | : Prevents others from using, making, or selling the invention without permission.

Slide 11: Patent | rights | are | territorial | Term: | 20 | years | from | date | of | filing | Aim | Encourage | industrial | development | Encourages | complete | disclosure | Prevents | duplication | Focus | on | commercial | relevance | Prevents | exploitation | of | researchers | Revenue | generation | Source | of | technical | information

Slide 12: Process | or | Method | Machine | or | Apparatus | Article | of | Manufacture | Composition | of | Matter | « | Chemical | Compounds | « | Physical | Mixtures | Improvements | of | Any | of | the | Above

Slide 13:

Slide 14: Novelty | mean | that | the | subject | matter | SHOULD | NOT | BE | PUBLISHED | IN | ANYWHERE | IN | THE | WORLD | PRIOR | TO | THE | PATENT | APPLICANT | INVENTING | IT | IN | PRIOR | PUBLIC | KNOWLEDGE | OR | PRIOR | PUBLIC | , | BUT | NOT | NECESSARILY | PATENTED | OR | PUBLISHED

Slide 15: USE, | SALE, | MANUFACTURE, | & | DEVELOPED | ANYWHERE | IN | THE | WORLD | CLAIMED | BEFORE | IN | A | PATENT | SPECIFICATION | ANYWHERE | IN | THE | WORLD

Slide 16: The | Novelty | Search | is | conducted | to | see | whether | the | invention | is | anticipated | by | any | prior | art

CSE332 Lecture 8.pptx (8 slides)

Slide 1: CSE332 | INDUSTRY ETHICS AND LEGAL ISSUES

Slide 2: Patents

Slide 3: Stages from filing to grant of a patent

Slide 4: Publication and Examination | Rule24-B:Examinationofapplication | Examination request for application: | 1. | 48 months | from filing date or | priority date | , whichever or | 6 months | from further application filing. | 2. Controller refers to Examiner | within 1 month | from publication or examination request receipt date whichever is | earlier. | 3. Examiner prepares report | in 1-3 months | .

Slide 5: A PROCEDURE FROM FILING TO | GRANT OF PATENT (Cont.)

Slide 6: IMPORTANT FEES IN DIFFERENT PROCEEDINGS

Slide 7: DUTIES OF PATENTEE | Maintenance of the patent | Pay renewal fees | Every year till the end of the term | Time limit: | before end of the | preceeding | year (6months extn. Available) | Working of the patents: | Inventions are to be worked in India on commercial scale

Slide 8: Litigation | vs | Licensing | If the objective is to stop a | competitor from offering a | p | roduct | that infringes one's patent. | If the objective is to obtain royalty | for the use of one's invention. | Alternatives To Litigation | Litigations are costly, timely, & disruptive | Alternative dispute resolutions (ADR) include | Mediation | & | Arbitration

CSE332 Lecture 9-10 (1).pptx (12 slides)

Slide 1: CSE332 | Lecture 8 and Lecture 9

Slide 2: Case Studies | Successful patent protection in renewable energy | Copyrights promoting creative digital content growth | Trademarks boosting brand identity for startups | IP disputes resolved through effective legal frameworks | Engineering innovations achieving global recognition | Lessons learned from intellectual property cases

Slide 3: Challenges in IP Management | Costly and time-consuming registration processes | Global variations in intellectual property laws | Enforcement difficulties across multiple jurisdictions | Balancing openness with competitive confidentiality | Navigating ethical considerations in patenting life | Ensuring accessibility while protecting ownership

Slide 4: Conclusion | Addressing social and engineering issues effectively | Conducting thorough analysis for innovative solutions | Registering claims to protect intellectual property | Promoting sustainable and ethical advancements globally | Encouraging collaboration for interdisciplinary success | Securing a better future through innovation

Slide 5: Conducting Analysis and Registering Claims | Identifying social and engineering issues | Conducting thorough analysis for solutions | Registering intellectual property protection claims | Critical steps for innovative project success | Balancing creativity and legal safeguarding rights | Promoting sustainable and ethical advancements

Slide 6: Introduction to Issues | Social issues impact technology adoption rates | Engineering challenges hinder development progress | Innovative ideas need effective legal safeguards | Analyzing | problems ensures impactful solutions | Registering claims protects intellectual property rights | Essential for fostering innovation and growth

Slide 7: Social Issues in Technology | Access inequality limits technology benefits | Cultural factors influence adoption practices | Privacy concerns create user trust barriers | Digital literacy varies across demographics | Ethical considerations shape public perceptions | Collaborative efforts address these challenges

Slide 8: Engineering Challenges Overview | Resource constraints limit design possibilities | Technological complexity increases implementation risks | Safety standards demand rigorous testing protocols | Scalability concerns affect future adaptability | Environmental impact guides material selection choices | Continuous improvement ensures optimal performance

Slide 9: Importance of Analysis | Thorough analysis identifies root problem causes | Stakeholder input ensures inclusive decision-making | Data-driven insights guide effective strategies | Scenario | modeling | anticipates potential outcomes | Structured approaches enhance solution feasibility | Collaboration fosters interdisciplinary innovation

Slide 10: Intellectual Property Basics | Protecting ideas safeguards competitive advantages | Types include patents, trademarks, copyrights | Filing claims requires detailed documentation processes | Legal protection prevents unauthorized usage instances | Encourages investment in research and development | Strengthens economic and creative ecosystems

Slide 11: Process of Registering Claims | Identify innovative aspects of your invention | Prepare detailed descriptions and technical drawings | Conduct searches for existing similar patents | File applications with relevant legal authorities | Respond to examination reports and objections | Maintain patents through timely fee payments

Slide 12: Benefits of IP Protection | Secures exclusive rights for commercial exploitation | Increases market value of protected innovations | Attracts funding for further developmental efforts | Prevents competitors from copying unique ideas | Strengthens brand recognition and customer trust | Supports long-term business sustainability goals

Unit 3: Government Funding & Startup Schemes

CSE332 Lecture 11 (1).pptx (15 slides)

Slide 1: CSE332 | Unit3:- Government Funding and startup schemes

Slide 2: Bank Loan for Startup Business in India | 1. MUDRA Loan (Pradhan Mantri MUDRA Yojana - PMMY) | ✓ | Loan Amount | : Up to | ₹10 lakh | ✓ | Eligibility | : Startups, MSMEs, and small businesses | ✓ | Types of Loans | : | Shishu | : Up to ₹50,000 (For new businesses) | Kishor | : ₹50,000 to ₹5 lakh (For growing businesses) | Tarun | : ₹5 lakh to ₹10 lakh (For established businesses) | ✓ | Collateral | : | No collateral required | ✓ | Interest Rate | : 8% to 12% (varies by bank) | ✓ | Where to Apply | : Public and private sector banks, NBFCs, microfinance institutions

Slide 3: Continue.... | 2. Stand-Up India Scheme | (For Women & SC/ST Entrepreneurs) | ✓ | Loan Amount | : ₹10 lakh to ₹1 crore | ✓ | Eligibility | : | At least | 51% ownership | by SC/ST or women entrepreneurs | Business must be in | manufacturing, services, trading, or agriculture | ✓ | Interest Rate | : Bank-dependent, competitive rates | ✓ | Where to Apply | : Banks and online via | www.standupmitra.in

Slide 4: Continue.... | 3. Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE) | ✓ | Loan Amount | : Up to | ₹2 crore | ✓ | Eligibility | : Startups & MSMEs in | manufacturing, service, and retail sectors | ✓ | Collateral | : | Collateral-free | for loans up to ₹2 crore | ✓ | Interest Rate | : 9% to 14% (bank-dependent) | ✓ | Where to Apply | : Public/private banks, NBFCs

Slide 5: Continue.... | 4. SIDBI Startup Loan | (Small Industries Development Bank of India) | ✓ | Loan Amount | : ₹10 lakh to ₹10 crore | ✓ | Eligibility | : Startups in innovation, technology, and MSME sectors | ✓ | Collateral | : Depends on the loan amount & business plan | ✓ | Interest Rate | : 9.5% onwards | ✓ | Where to Apply | : | www.sidbi.in | or partner banks

Slide 6: Continue.... | 5. Bank-Specific Startup Loans | Many banks offer | customized startup loans | : | SBI Startup Loan | – Business loans for MSMEs and startups | HDFC Bank Business Growth Loan | – Unsecured loans for startups | ICICI Bank Business Loan | – Loans for working capital and expansion | Axis Bank Business Loan | – Quick approval loans for small businesses

Slide 7: How to Apply for a Startup loan? | Prepare Documents | : | Business plan | Financial projections | GST registration, PAN, Aadhaar | Bank statements (last 6 months) | Choose a Loan Scheme | (MUDRA, Stand-Up India, CGTMSE, etc.) | Apply Online or Visit a Bank | Submit Documents & Get Approval

Slide 8: What are startups | Startups often operate with limited resources and rely on funding from investors, venture capitalists, or crowdfunding. | They typically focus on rapid growth and scalability to achieve success in a competitive market. | Many startups begin as small teams and evolve into larger companies if they succeed. | While some startups achieve great success, many face challenges and risks that can lead to failure.

Slide 9: India (DPIIT - Startup India) | The | Department for Promotion of Industry and Internal Trade (DPIIT) | defines a startup as: | A company that has been in existence for | less than 10 years | from its date of incorporation. | Its annual turnover does not exceed | INR 100 crore | in any financial year. | It is working towards | innovation, development, or improvement of products, processes, or services | or has a | scalable business model | with high potential for employment and wealth creation. | It should | not be formed by splitting or reconstructing an existing business | .

Slide 10: Government Startup Schemes in India & Worldwide(Resources) |  India (Startup India & Other Schemes) | Startup India Initiative | – Provides tax exemptions, funding support, and ease of business registration. | MUDRA Loan Scheme | – Offers collateral-free loans up to ₹10 lakh for small startups. | Stand-Up India | – Supports SC/ST and women entrepreneurs with loans between ₹10 lakh – ₹1 crore.

Slide 11: Continue.. | 4. SIDBI Fund of Funds | – ₹10,000 crore fund to invest in startups through venture capital firms. | 5. Atal Innovation Mission (AIM) | – Supports innovation and incubation centers in schools &

universities. | 6. E-Marketplace | – Allows startups to sell products/services to the government easily. | 7. SEED Fund Scheme | – Provides financial assistance of up to ₹50 lakh for early-stage startups

Slide 12: Benefits of Startup India Scheme | 1. Tax Benefits | Tax Holiday for 3 Years | – Startups registered under | DPIIT | (Department for Promotion of Industry and Internal Trade) get a | 100% tax exemption | for the first | 3 consecutive years | (under Section 80-IAC of the Income Tax Act). | Exemption from Tax | – Startups are exempt from the | Tax | (Section 56(2)(| vii |)) on investments received from investors. | Capital Gains Tax Exemption | – Startups can claim | exemptions on long-term capital gains | if the amount is reinvested in a fund approved by the government.

Slide 13: Benefits of Startup India Scheme | 2. Easy Compliance & Self-Certification | Startups can | self-certify | compliance for | 9 labor and environmental laws | , reducing the regulatory burden. | Fewer inspections in the initial years. | 3. Simplified Company Registration | Startups can | register online | via the | Startup India Portal | with minimal documentation. | 4. Funding Support | Fund of Funds for Startups (FFS) | – ₹10,000 crore fund managed by | SIDBI | to support venture capital firms investing in startups. | Credit Guarantee Scheme | – Provides collateral-free loans to startups.

Slide 14: Benefits of Startup India Scheme | 5. Fast-Track Patent & IPR Protection | Startups get | 80% rebate | on patent filing fees and | 50% rebate | on trademark fees. | Fast-track patent examination | and | free legal assistance | for intellectual property rights. | 6. Government Tenders & Procurement Benefits | Startups | do not need prior experience | to apply for | government tenders | . | They get | priority in public procurement | from PSUs and government departments.

Slide 15: Benefits of Startup India Scheme | 7. Incubation & Networking Opportunities | Startup India Hub | connects entrepreneurs with mentors, investors, and incubators. | Access to | Atal Incubation Centers (AICs) | and | Startup Accelerators | . | 8. Easier Exit for Startups | Startups can | wind up within 90 days | under the | Insolvency and Bankruptcy Code, 2016 | instead of the usual 180 days.

CSE332 Lecture 12.pptx (14 slides)

Slide 1: CSE332 | Unit3:- Government Funding and | startup | schemes | Topics: | Start-up India, 10000 startups -A NASSCOM | Initiative

Slide 2: Key Features of 10,000 | Startups | Startup | Incubation & Acceleration Programs | Funding Support via Investor Connects | Mentorship from Industry Experts | Global Expansion & Market Access | Corporate & Government Partnerships | Example | : | Razorpay | – Grew rapidly through | startup | mentorship and funding.

Slide 3: Over | 7,500 | startups | incubated | 300 | + | startups | scaled globally | 50+ Accelerator programs conducted | \$4 billion+ funding raised | Example: | Unacademy | – Benefited from NASSCOM's ecosystem and became a leading | EdTech | platform. | Impact of 10,000 | Startups

Slide 4: Challenges | Faced by | Startups | Funding | & Investment | Barriers | Market Competition & Scalability Issues | Regulatory & Compliance Challenges | Talent Acquisition & Retention | Example: Many AI-based | startups | struggle with data privacy regulations and funding gaps.

Slide 5: Future of 10,000 | Startups | Initiative | Expanding to Tier-2 & Tier-3 Cities | Focus on AI, | DeepTech | , and Sustainability | More Collaboration with Government & Enterprises | Scaling Indian | Startups | for Global Markets | Example: Digit Insurance – A | startup | that leveraged NASSCOM's network to expand in the | InsurTech | sector | .

Slide 6: 10,000 Startups is driving India's startup | ecosystem | Provides funding, mentorship, and global opportunities | Helps | startups overcome challenges & scale efficiently | Continuous support for emerging technologies & | entrepreneurs | Final | Thought: India's future as a global startup hub depends on initiatives like NASSCOM's 10,000 Startups! | Conclusion & Key Takeaways

Slide 7: Start-up India | , launched on | January 16, 2016 | , | Aims | to | promote entrepreneurship, innovation, and job creation | 3. Through | financial support, legal simplifications, and | mentorship | . | Examples : Benefiting | startups like | BYJU'S, | Zomato | , and | Paytm | . | Start-up India

Slide 8: Benefits & Incentives for Startups | Simplified | Compliance: | Self-certification for labor and environmental laws. | Tax Exemptions: | 3-year income tax holiday for eligible startups. | Funding Support: |

₹10,000 crore | Fund of Funds | for startups. | Fast-Track Patent Process: | 80% reduction in patent filing fees. | Example: | Ather | Energy (EV Startup) | leveraged government incentives to scale its business.

Slide 9: Eligibility & Registration Process | Must be incorporated as a Private Limited Company, LLP, or Partnership Firm | Should be less than 10 years old | Annual turnover should not exceed ₹100 crore | Must be working on innovation, development, or improvement of products/services | Example | : | Zerodha | (Fintech Startup) registered under Start-up India and grew into a market leader.

Slide 10: Funding | & Investment | Opportunities | Fund | of Funds Scheme (FFS): | Invests in startups through SEBI-registered VC firms | . | Start-up India Seed Fund Scheme (SISFS): | ₹945 crore to support early-stage startups. | Government | Grants & Incubation Centers: | Provides infrastructure, networking, and financial support. | Example | : | Cure.fit | (| HealthTech | Startup) | received funding through various government and private investors.

Slide 11: Challenges Faced by Indian Startups | Funding Constraints: | Many startups struggle to secure investment | . | Regulatory Hurdles: | Compliance and legal formalities take time. | Market Competition: | High competition from global and local players. | Scaling Issues: | Difficulties in expanding due to operational challenges. | Example: | Kirana | King (Retail Startup) | overcame scaling challenges through digital transformation.

Slide 12: Impact & Future of Start-up India | Over | 100,000 startups recognized | by the DPIIT (Department for Promotion of Industry and Internal Trade). | India is the 3rd largest startup ecosystem | globally. | Focus Areas for Growth: | DeepTech | , AI, Green Energy, | FinTech | , | Agritech | . | Government's Next Steps: | Improving ease of doing business, increasing startup funding, and promoting exports | . | Example | : | Nykaa | (Beauty Startup) | started as a small business and became a publicly traded company.

Slide 13: . | 10000 | startups | -A NASSCOM Initiative

Slide 14: Introduction | & Launch | – NASSCOM initiated in | 2013 | for startups. | Objective & Vision | – Support, mentor, and scale | 10,000 startups | . | Key Focus Areas | – AI, | FinTech | , | HealthTech | , SaaS, | EdTech | . | Startup Support | – Funding, mentorship, global expansion, acceleration. | Impact Created | – Thousands incubated, billions in funding. | Example | Success – | Freshworks | scaled, India's first SaaS unicorn | .

CSE332 Lecture 13.pptx (13 slides)

Slide 1:

Slide 2: Key Differences: STP vs. SEZ | A side-by-side comparison of the two schemes: | • **Sector Focus:** STPs specialize in IT/Software, while SEZs cater to multiple sectors. | • **Tax Holiday:** STPs provide a 10-year benefit, compared to 15 years for SEZs. | • **Location:** STPs are typically situated in dedicated parks, whereas SEZs span larger geographic zones.

Slide 3: Driving Growth & Employment | Together, STPs and SEZs have fueled significant economic growth: | • STPs have generated approximately 4.5 million jobs in the IT sector. | • SEZs have created over 2 million direct jobs as of 2023. | • Combined, these initiatives support exports valued at around ₹20 lakh crore.

Slide 4: Summary & Next Steps | In summary, STPs and SEZs are foundational pillars of India's export growth and economic progress. | They provide a robust platform for future initiatives such as 'Make in India' and support for startups. | Call to Action: | • Explore case studies (e.g., Wipro's success within an STP). | • Stay updated on forthcoming policy changes in 2024.

Slide 5: Data Sources | • STPI Annual Report (2022-23) | • SEZ India Portal (2023) | • Ministry of Commerce & Industry | Visual Suggestion: Include government logos (STPI, SEZ, MeitY) to reinforce credibility.

Slide 6: India's Export Promotion Schemes: STPs & SEZs | Boosting Global Competitiveness

Slide 7: Real-World Impact of STPs & SEZs | • Infosys, leveraging the STP model, achieved \$15.4B in exports during 2022-23. | • SEZs contributed to 20% of India's total exports in the same period. | These figures underscore how these schemes are key drivers of global trade.

Slide 8: STP Scheme: Digital Export Engine | STP Registration Flowchart | Start | Begin the STP registration process. | Eligibility Check | Ensure the unit qualifies for STP registration (e.g., software development, IT services, etc.). | Prepare Documents | Gather required documents: | Application form (Form A). | Project report. | Memorandum of Association (MOA) and Articles of Association (AOA). | Board resolution for STP registration. | PAN card, GST registration, and other legal documents. | Online Application | Submit the application on the STPI portal (| <https://www.stpi.in> |). | Application Review | STPI reviews the application and documents for completeness and accuracy.

Slide 9: Clarifications/Corrections (if any) | If STPI requests additional information or corrections, provide the required details. | Site Inspection | STPI may conduct a site inspection of the proposed unit. | Approval/Rejection | STPI approves or rejects the application based on the review and inspection. | Approval Granted | If approved, the unit receives an STP registration certificate. | Commence Operations | Start operations under the STP scheme and avail benefits (e.g., tax exemptions, customs duty benefits). | Compliance | Submit monthly/quarterly/annual returns and reports to STPI as required. | End | Process complete.

Slide 10: STP Scheme: Digital Export Engine | Since its inception in 1991 by the Ministry of Electronics & IT, the STP scheme has transformed India's export landscape. | Key Features: | • Tax holidays and a streamlined single-window clearance process | • 100% FDI allowed in IT/ITES sectors | • Over 600 registered units as of 2023 | • Focus on software development, R&D, and innovation

Slide 11: Why STPs? Key Incentives | STPs offer a suite of attractive benefits that have made them a cornerstone of India's export strategy: | • 10-year income tax exemption | • Duty-free imports for advanced technology | • Simplified compliance procedures | • Robust cloud infrastructure support | • Enhanced export revenue retention (contributing ~45% to IT exports)

Slide 12: SEZ Scheme: Trade & Investment Hubs | Launched in 2005 under the SEZ Act, Special Economic Zones have emerged as pivotal trade and investment hubs. | Highlights: | • Over 270 operational SEZs as of 2023 | • Significant tax breaks and GST exemptions | • Boost to export-oriented manufacturing and services | • Strategic clusters in states like Gujarat and Tamil Nadu

Slide 13: SEZ Advantages for Exporters | SEZs offer compelling advantages to exporters, including: | • A 15-year corporate tax holiday | • Duty-free import and export privileges | • An FDI framework that requires no additional licensing | • Simplified labor laws | In 2022-23, SEZ exports reached an impressive ₹7.5 lakh crore.

Unit 4: Startup in IT & Business Plans

CSE332 Lecture 15.pptx (15 slides)

Slide 1: CSE332 | Unit 4:- Startup In IT

Slide 2: General Company Description | Company Name | Founded | Location | Business Structure: | Sole proprietorship, LLC, or Private Limited. | Founders & Team: | Names and expertise.

Slide 3: Mission Statement | "To deliver innovative IT solutions that enhance business efficiency, security, and scalability, enabling digital transformation for businesses of all sizes."

Slide 4: Vision Statement | "To become a leading IT solutions provider known for innovation, reliability, and customer-centric technology solutions."

Slide 5: Market Analysis | Industry Growth | Target Market Needs: | Demand for AI, cloud computing, and cybersecurity. | Businesses seeking digital transformation. | Opportunity

Slide 6: Core Values | Innovation | – Continuous tech advancements. | Customer-Centric Approach | – Prioritizing client needs. | Integrity & Security | – Ensuring data protection. | Scalability & Efficiency | – Providing adaptable solutions.

Slide 7: Contact Information | Company Name | Website | Email | Phone | Address

Slide 8: IT Startup Business Plan | Executive Summary & General Company Description

Slide 9: Executive Summary

Slide 10: Why Business Plan is important | – Helps secure funding from banks or venture capitalists. | – Acts as a strategic guide for business operations. | – Helps in planning for market challenges. | – Helps track growth and measure success.

Slide 11: Executive Summary | Introduction: | Overview of the startup. | Industry: | IT sector with a focus on specific niche. | Business Model: | Subscription-based, SaaS, consulting, or custom solutions. | Target Market: | Small businesses, enterprises, healthcare, fintech, etc.

Slide 12: Key Offerings | Services/Products: | Custom software development | AI-driven automation | Cloud solutions | IT consulting | Cybersecurity solutions

Slide 13: Goals & Objectives | Launch MVP within [timeframe]. | Acquire [X] clients in the first year. | Expand team and operations by [timeframe]. | Achieve revenue growth of [X%] annually.

Slide 14: Financial Overview | Initial Investment: | \$X (for development, marketing, and operations). | Revenue Streams: | Subscription-based services | Software licensing | Consulting fees | Projected ROI: | Within [X] years.

Slide 15: Competitive Advantage | Innovation: | AI-driven analytics, automation. | Cost-Effective Solutions: | Affordable IT services. | Expert Team: | Skilled professionals in IT and software development. | Scalability: | Solutions adaptable to business growth.

CSE332 Lecture 16.pptx (18 slides)

Slide 1: CSE332 | Industrial | Ethics | and | Legal | Issues | Product Based and Service Based IT Companies , Marketing plan ,Operational Plan , Management and Organization

Slide 2: Operational Plan | An | Operational Plan | is a highly detailed | plan | that provides a clear picture of how a team, section or department will contribute to the achievement of the | organisation's goals.

Slide 3: Operational Plan | Defines company processes, workflows, and execution. | Ensures efficiency, productivity, and goal alignment. | Covers resource management and daily operations. | Includes risk assessment and contingency planning. | Optimizes workforce for maximum output efficiency. | Monitors performance using key operational metrics.

Slide 4: Business Process Management | Standardize workflows for smooth project execution. | Utilize agile or waterfall development methodologies. | Implement automation for increased operational efficiency. | Ensure seamless coordination across business departments. | Regularly review and refine existing processes. | Enhance customer satisfaction through better service.

Slide 5: Resource Allocation Strategy | Assign resources based on project requirements. | Optimize manpower for cost-effective utilization. | Leverage cloud computing for scalable infrastructure. | Maintain budget control for operational sustainability. | Ensure backup systems for disaster recovery. | Reduce downtime through proactive resource planning.

Slide 6: Technology and Infrastructure | Invest in secure and scalable systems. | Ensure IT infrastructure supports business needs. | Regularly upgrade hardware and software tools. | Implement cybersecurity measures for data protection. | Leverage AI and automation for efficiency. | Monitor system performance and prevent failures.

Slide 7: Risk Management and Compliance | Identify potential risks in business operations. | Develop contingency plans for risk mitigation. | Ensure compliance with legal IT regulations. | Conduct regular audits for process improvements. | Implement data protection and cybersecurity policies. | Train employees on security and compliance.

Slide 8: Performance Monitoring and Optimization | Track operational KPIs for business growth. | Evaluate team efficiency and project timelines. | Use data analytics for process improvements. | Regularly update workflows for better performance. | Encourage innovation for operational excellence. | Review strategies and optimize for success.

Slide 9: Management and Organization | Management Team :- | Founders & core executives. | Industry experience overview. | Leadership team expertise. | Key responsibilities assigned. | Growth & vision alignment. | Advisors & mentors onboard.

Slide 10: Management and Organization | Organizational Structure:- | Startup type & ownership | Hierarchical team structure | Key departments overview | Roles & responsibilities defined | Operational workflow summary | Decision-making authority

Slide 11: Product Based Companies | Develop and sell software-based products. | Invest heavily in R&D and innovation. | Example companies: Microsoft, Google, Adobe. | Earn revenue from licensing and subscriptions. | Focus on scalability, automation, and efficiency. | Customer loyalty depends on product quality.

Slide 12: Service Based Companies | Offer IT services and consulting globally. | Work on client-specific software solutions. | Example companies: TCS, Infosys, Wipro, Accenture. | Revenue depends on contracts and projects. | Skilled workforce is crucial for success. | Customized solutions meet client requirements.

Slide 13: Marketing Plan | A | marketing plan | is a comprehensive document or blueprint that outlines a | business advertising and | marketing | efforts | for the coming year.

Slide 14: Marketing Plan | Define goals, audience, and key strategies. | Focus on digital and offline presence. | Increase brand awareness and customer engagement. | Monitor trends and | analyze | competitors' strategies. | Measure success using key performance indicators.

Slide 15: Marketing Plan- Target Market Analysis | Identify industries needing IT-based solutions. | Analyze | customer expectations. | Segment audience based on demographics and needs. | Create buyer personas for personalized marketing. | Understand market trends and emerging demands. | Align marketing strategies with audience | behavior | .

Slide 16: Digital Marketing Strategy | Leverage paid ads for quick conversions. | Use email marketing for lead nurturing. | Engage customers through social media platforms. | Track and optimize campaigns continuously.

Slide 17: Customer Engagement and Retention | Provide excellent support for client satisfaction. | Use AI-driven chatbots for quick responses. | Personalized communication enhances brand trust. | Collect feedback to improve service quality. | Build long-term relationships for brand loyalty.

Slide 18: Performance Measurement and Optimization | Track key metrics using analytics tools. | Monitor website traffic and conversion rates. | Analyze | marketing spend and ROI performance. | Refine strategies based on customer feedback. | Adjust campaigns for better market impact. | Stay updated with industry trends regularly | .

CSE332 Lecture 17.pptx (32 slides)

Slide 1: Startup | in | IT | UNIT | 4

Slide 2: For Each Item: | Determine expected cash receipt timing. | Track when expenses need payments. | Opening Day Balance Sheet: | Fundamental financial report for businesses. | Essential for reporting and management. | Break-Even Analysis: | Dividing line between loss and profit. | Sales level needed to break even.

Slide 3: APPENDICES | The | appendices | contain | letters | and | charts | with | important | data | that | not | only support the book's | arguments, | but | also | open | new | ways | for | future | investigations.

Slide 4: AP | P | ENDIC | E | S | Include | details | and | studies | used | in | your | business | plan; | for | example: | Brochures | and | advertising | materials | Industry | studies | Maps | and | photos | of | location | Magazine | or | other | articles

Slide 5: Detailed | lists | of | equipment | owned | or | to | be | purchased | Copies | of | leases and | contracts | Letters | of | support | from | future | customers | Any | other | materials | needed | to | support | the | assumptions | in | this | plan | Market | research | studies | List | of assets | available | as | collateral | for | a | loan

Slide 6: Refining | the | Plan | The generic | business | plan | presented | above | should | be | modified | to | suit | your | specific type | of | business | and the | audience | for | which | the plan is | written

Slide 7: Refining | the | Plan | For | Raising | C | apital | For | Bankers | Amount of Loan: | Clearly state the loan amount requested. | How Funds Will Be Used: | Specify exactly how loan funds will be allocated. | What This Will Accomplish: | Explain how funds strengthen the business. | Requested Repayment Terms: | Outline the repayment schedule (years). | Collateral Offered: | List the collateral securing the loan.

Slide 8: Refining | the | Plan | For | Investors | Funds Needed Short-Term: | capital for growth opportunities. | Funds Needed in 2-5 Years: | Scaling and market expansion goals. | How Funds Will Be Used: | Product development, marketing, operational expansion. | Estimated Return on Investment: | Projected high returns for investors.

Slide 9: Refining | the | Plan | For | Type | of | Business | Manufacturing | Planned Production Levels: | Set targets for product output. | Production Cost Comparison: | Compare direct/indirect costs to industry averages. | Product Line Pricing: | Define prices for each product. | Inventory Management & New Products: | Procedures and upcoming product developments.

Slide 10: Refining | the | Plan | Service | Businesses | Key Competitive Factors: | Pricing, quality, and service flexibility. | Pricing Methods: | market-based, or competitive pricing. | Production Management System: | Efficient processes, timely service delivery. | Quality Control Procedures: | Industry standards, consistent service quality. | Labor Productivity Measurement: | Output per worker, efficiency tracking. | Subcontracting, Credit, and Client Strategy: | Subcontracting, payments, and client retention.

Slide 11: STARTUP | STORY | FLIPKART | HISTORY | Back | in | 2007, | when | Flip | kart | was | launched, | Indian | e- | commerce | industry | was | taking | its | beginner | steps. | The | company | is | registered | in | Singapore, | but | their | headquarters | are | in | the city | of | Bangalore, | India.

Slide 12: Personal | Financial | Statement | A | personal | financial | statement | is | a | document | or | spreadsheet | outlining | an | individual's | financial | position | at | a | given point | in | time.

Slide 13: Founders | Sachin | Bansal | and | Binny | Bansal | , | who | were | working | f | or | Amazon.com | had | an | idea | to | start | an | e-commerce | company | in | India. | Both | of | them | are | alumni | of | IIT, | Delhi | and | are | native | of | Chandigarh, | India. | They | left | their | jobs | in | Amazon | to | start | their | own | business.

Slide 14: How | it | Started? | Flipkart | began | selling | books | to | begin | with. | It | soon | expanded | and | began | offering | a wide | variety | of | goods

Slide 15: Funding | History | This soon increased | to | 200 million | Indian rupees the | following | year. | Their | last | round | of | Fundraising | had | increased | their | value | to | \$ | 15 | billion.

Slide 16: Evolution | Flipkart revolutionized online shopping in India by introducing Cash on Delivery (COD) and card payment on delivery, making it more accessible. COD became a standard feature across Indian e-commerce platforms after Flipkart's pioneering implementation.

Slide 17: Evolution | Flipkart solved delivery issues by creating its own system to ensure timely shipments. This helped them deliver orders quickly and reliably.

Slide 18: Acquisitions | Flipkart expanded its market presence by acquiring companies like Myntra and | LetsBuy | , transforming from a small book retailer to India's largest e-commerce platform. Its journey has inspired startups and revolutionized the Indian e-commerce industry, breaking stereotypes along the way.

Slide 19: STARTUP | STORY | Make | My | Trip | FOUNDER | Deep | Kalra | is the | proud | founder | of | Gurgaon | based | – | MakeMyTrip.com.

Slide 20: STARTUP | STORY | Make | My | Trip | FOUNDER | How | did | his | journey | begin? | Deep Kalra began his career after completing his Masters at IIM-A and worked at companies like GE Capital and ABN AMRO Bank. In 1995, he took a big risk by leaving his stable job to join AMF Bowling, an American company looking to set up bowling alleys and billiard halls in India.

Slide 21: STARTUP | STORY | Make | My | Trip | FOUNDER | How | did | his | journey | begin? | Deep Kalra worked hard and managed to open over 200 bowling lanes, mostly in small centers. However, the venture wasn't his own, and he had a distant boss in America who didn't provide the proper guidance or support.

Slide 22: STARTUP | STORY | Make | My | Trip | FOUNDER | How | did his | journey | begin? | Evidently, | for | obvious | reasons | this | did | not | appeal | to | him | and | he | also | realized | that | they | were | all | not | worth | his | talent. | He | needed | something | more | to | showcase | the | same!

Slide 23: Personal | Financial | Statement | Assets and liabilities outside business. | Personal net worth for each owner. | Key for business financing clarity. | Owners may use personal assets. | Shows available assets for business. | Investors and bankers require details.

Slide 24: STARTUP | STORY | Make | My | Trip | FOUNDER | How | did | his | journey | begin? | Deep Kalra noticed the growth of the internet and quit his job to simplify the messy online travel booking industry. He founded MakeMyTrip, transforming travel bookings and quickly making it one of India's leading travel platforms.

Slide 25: STARTUP | STORY | Achievements | Chairperson | of | the | NASSCOM | Internet | Working | Group | Member | of the | Executive | Council | of | NASSCOM | Member | of | CII's | Tourism | sub-committee | Awarded | as the | "Best | Travel | Portal | India" | by | World | Travel | Awards | (2014, | 2013) | Awarded | as | the | "E-tailer | of | the | Year" | by | ET | Retail | Awards | (2014, | 2013) | Awarded | as | the | "Best | Online | Travel | Service | Firm" | by | Times | Travel | Honours | (2011) | Listed | as | "Ten | Best | Companies | to | Work | For | in | India" | by | Great | Place | to | Work | Institute | (2013, | 2012, | 2011, | 2010)

Slide 26: Thank you

Slide 27: Startup | Expenses | and | Capitalization | Startup Expenses: | List of individual items required to start the business. | Includes every necessary resource for business launch. | Capitalization: | Total required investment to launch the business. | Represents the full cost of resources planned to be leveraged.

Slide 28: Entrepreneurs Should Clearly List: | Startup Expenses: | Every cost associated with getting the business operational. | Associated Costs: | Each item's specific cost breakdown. | Unusual or Expensive Items: | Brief explanation of capital-intensive or expensive items. | Highlight any large upfront costs for critical items.

Slide 29: Startup | Expenses | and | Capitalization

Slide 30: Startup | Expenses | and | Capitalization

Slide 31: Financial | Plan | Financial | planning | is the | task | of | determining | how a | business | will | afford | to | achieve | its | strategic | goals | and | objectives.

Slide 32: Financial | Plan | 12 Month Profit & Loss Statement: | Put numbers together for profitability. | Helps determine the path to success. | Three-Year Profit Projection (Optional): | Extends forecast beyond the first year. | Helps plan for long-term growth. | Projected Cash Flow: | Plan how much cash you need. | Includes preliminary expenses and reserves.

Unit 5: IT & ITES Industry

CSE332 Lecture 18.pptx (13 slides)

Slide 1: Introduction to IT and ITES Industry | Product-Based vs. Services-Based Companies, NASSCOM, STPI, and Global IT Projects

Slide 2: Impact of Emerging Tech | - AI: Intelligent automation | - IoT: Smart devices | - Cloud: Scalable solutions | - Blockchain: Secure transactions | - 5G: Faster connectivity | - Quantum computing: Future potential

Slide 3: Govt Initiatives for IT | - Digital India Program | - Make in India | - IT SEZs & incentives | - Startup India support | - NASSCOM's skill programs | - Global collaboration efforts

Slide 4: Successful IT Projects | - Aadhaar (India's digital ID) | - Google's AI-driven search | - Amazon AWS cloud expansion | - Tesla's self-driving tech | - Paytm's fintech revolution | - OpenAI's ChatGPT & AI models

Slide 5: Future of IT & ITES | - AI & ML growth continues | - Cybersecurity challenges increase | - More automation & robotics | - Expansion of remote work | - Tech-driven economies | - Global IT talent demand rises

Slide 6: What is IT and ITES? | - IT: Information Technology | - ITES: IT-Enabled Services | - IT focuses on software & hardware | - ITES includes BPO, KPO, and LPO | - Growth driven by digitalization | - Key role in global economy

Slide 7: Product vs. Services Companies | - Product: Develop own software/hardware | - Examples: Microsoft, Google | - Service: IT solutions for clients | - Examples: TCS, Infosys | - Product: Higher R&D investment | - Service: Client-based revenue model

Slide 8: NASSCOM: Industry Body | - National Association of Software & Services | - Represents India's IT-BPM industry | - Helps policy-making & innovation | - Focus on skill development | - Encourages global collaborations | - Organizes IT summits & reports

Slide 9: STPI: Software Technology Parks | - Software Technology Parks of India | - Supports software exports | - Provides infrastructure & incubation | - Tax benefits for IT companies | - Promotes startups & innovation | - Key to India's IT growth

Slide 10: Latest IT Projects | - AI & Machine Learning in Healthcare | - Cloud computing transformations | - Blockchain for secure transactions | - 5G technology advancements | - Smart cities & IoT adoption | - Cybersecurity & data privacy initiatives

Slide 11: Conclusion | - IT & ITES drive innovation | - Product vs. Service-based firms | - NASSCOM & STPI key enablers | - Emerging global IT projects | - Future: AI, Cloud, and Cybersecurity | - IT revolutionizing global industries

Slide 12: IT & ITES Market Trends | - Rapid cloud adoption | - AI & automation growth | - Increased cybersecurity focus | - Expanding remote workforce | - Rise of data-driven decisions | - Growing demand for IT talent

Slide 13: Major IT Hubs | - India: Bengaluru, Hyderabad, Pune | - USA: Silicon Valley, Seattle | - Europe: London, Berlin | - China: Beijing, Shenzhen | - Emerging hubs: Vietnam, Poland | - IT clusters drive innovation

CSE332 Lecture 19.pptx (20 slides)

Slide 1: Unit 5 | Case study of an IT industry (Product based and Services based) | Recent technology advancement | Current affairs related with the IT industry | Diversity in the Workforce

Slide 2: Service-based IT Companies: | Tata Consultancy Services (TCS) | What They Do: | TCS provides IT services like software development, consulting, cloud services, and IT support to businesses across the world. | How They Make Money: | By offering IT consulting and development services to businesses. | Example of Services: | Custom software development, cloud migration, IT infrastructure management.

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Slide 18: Product-based IT Companies: | Apple | What They Do: | Apple makes and sells products like | iPhones | , | MacBooks | , | iPads | , and software like | iOS | . | How They Make Money: | By selling their devices and software. They also earn from their App Store and subscription services like | iCloud | and Apple Music. | Example of Products: | iPhone | , | MacBook | , Apple Watch, | iPad | , and software like | iOS | .

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Slide 20: Sony | What They Do: | Sony develops electronic products like PlayStation consoles and also produces entertainment content (movies, music). | How They Make Money: | Through sales of gaming consoles, media content, and consumer electronics. | Example of Products: | PlayStation gaming consoles, | Xperia | phones, Sony | Bravia | TVs.

Extended_IT_and_ITES_Industry_Presentation.pptx (13 slides)

Slide 1: Introduction to IT and ITES Industry | Product-Based vs. Services-Based Companies, NASSCOM, STPI, and Global IT Projects

Slide 2: Impact of Emerging Tech | - AI: Intelligent automation | - IoT: Smart devices | - Cloud: Scalable solutions | - Blockchain: Secure transactions | - 5G: Faster connectivity | - Quantum computing: Future potential

Slide 3: Govt Initiatives for IT | - Digital India Program | - Make in India | - IT SEZs & incentives | - Startup India support | - NASSCOM's skill programs | - Global collaboration efforts

Slide 4: Successful IT Projects | - Aadhaar (India's digital ID) | - Google's AI-driven search | - Amazon AWS cloud expansion | - Tesla's self-driving tech | - Paytm's fintech revolution | - OpenAI's ChatGPT & AI models

Slide 5: Future of IT & ITES | - AI & ML growth continues | - Cybersecurity challenges increase | - More automation & robotics | - Expansion of remote work | - Tech-driven economies | - Global IT talent demand rises

Slide 6: What is IT and ITES? | - IT: Information Technology | - ITES: IT-Enabled Services | - IT focuses on software & hardware | - ITES includes BPO, KPO, and LPO | - Growth driven by digitalization | - Key role in global economy

Slide 7: Product vs. Services Companies | - Product: Develop own software/hardware | - Examples: Microsoft, Google | - Service: IT solutions for clients | - Examples: TCS, Infosys | - Product: Higher R&D investment | - Service: Client-based revenue model

Slide 8: NASSCOM: Industry Body | - National Association of Software & Services | - Represents India's IT-BPM industry | - Helps policy-making & innovation | - Focus on skill development | - Encourages global collaborations | - Organizes IT summits & reports

Slide 9: STPI: Software Technology Parks | - Software Technology Parks of India | - Supports software exports | - Provides infrastructure & incubation | - Tax benefits for IT companies | - Promotes startups & innovation | - Key to India's IT growth

Slide 10: Latest IT Projects | - AI & Machine Learning in Healthcare | - Cloud computing transformations | - Blockchain for secure transactions | - 5G technology advancements | - Smart cities & IoT adoption | - Cybersecurity & data privacy initiatives

Slide 11: Conclusion | - IT & ITES drive innovation | - Product vs. Service-based firms | - NASSCOM & STPI key enablers | - Emerging global IT projects | - Future: AI, Cloud, and Cybersecurity | - IT revolutionizing global industries

Slide 12: IT & ITES Market Trends | - Rapid cloud adoption | - AI & automation growth | - Increased cybersecurity focus | - Expanding remote workforce | - Rise of data-driven decisions | - Growing demand for IT talent

Slide 13: Major IT Hubs | - India: Bengaluru, Hyderabad, Pune | - USA: Silicon Valley, Seattle | - Europe: London, Berlin | - China: Beijing, Shenzhen | - Emerging hubs: Vietnam, Poland | - IT clusters drive innovation

Unit 6: Cyber Law & Information Security

CSE332 Lecture 18.pptx (13 slides)

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CSE332 Lecture 20.pptx (19 slides)

Slide 1: CSE332 | INDUSTRY ETHICS AND LEGAL ISSUES | Ethical and Professional issues in Information Security

Slide 2: Organizations face significant legal and ethical responsibilities regarding information security | . | Failure to comply with security laws and ethical standards can result in legal penalties, financial losses, and reputational damage | . | Organizational Liability and the Need for Counsel

Slide 3: Organizational Liability in Information Security | Organizations can be held liable for: | Data Breaches | – If customer or employee data is compromised due to negligence. | Non-Compliance with Regulations | – Violating laws like GDPR, HIPAA, or PCI DSS can result in heavy fines. | Cybersecurity Negligence | – Failure to implement security measures can lead to legal action. | Intellectual Property Violations | – Unauthorized use of copyrighted or patented technology. | Insider Threats and Employee Misconduct | – Misuse of company data by employees leading to legal liability.

Slide 4: Need for Legal Counsel in Cybersecurity | Organizations require legal counsel to: | Ensure Compliance | – Guide businesses on adhering to global cybersecurity regulations. | Manage Data Protection Policies | – Develop legal policies for handling customer and employee data. | Handle Cybersecurity Incidents | – Provide legal defense in case of breaches or cyberattacks. | Draft Contracts and Agreements | – Ensure secure and legally binding terms in business deals. | Risk Assessment and Mitigation | – Identify potential legal risks and prevent liabilities.

Slide 5: Examples of Organizational Ethics and Liability in Information Security | Facebook-Cambridge | Analytica | Scandal (Ethical and Legal Violations in Data Privacy) | Equifax Data Breach (Failure to Ensure Ethical Cybersecurity Practices) | Uber Data Breach Cover-Up (Lack of Transparency and Ethical Conduct) | Wells Fargo Fake Accounts Scandal (Unethical Employee Behavior and Corporate Accountability) | Yahoo Data Breach (Corporate Negligence in Cybersecurity and Legal Repercussions)

Slide 6: What happens without ethical employee conduct? | How does unethical behavior harm organizations? | What legal risks arise from misconduct? | Why is customer trust easily lost? | Can organizations face penalties for employees? | How does non-compliance affect reputation? | Answer the following questions

Slide 7: 1 | . Definition | Policy | – Internal guidelines set by organizations to govern behavior. | Law | – Legally enforceable rules established by governments. | 2. Purpose | Policy | – Ensures consistency, security, and ethical conduct within an organization. | Law | – Maintains order, protects rights, and enforces justice in society. | 3. Enforcement | Policy | – Enforced internally by management or HR departments. | Law | – Enforced by government authorities and courts. | Policy Versus Law

Slide 8: 4. Consequences of Violation | Policy | – Can lead to warnings, termination, or internal penalties. | Law | – Can result in legal action, fines, or imprisonment. | 5. Flexibility | Policy | – Can be modified by the organization as needed. | Law | – Requires legal procedures or legislative approval to change. | 6. Scope | Policy | – Applies only within an organization or institution. | Law | – Applies to everyone within a jurisdiction.

Slide 9: Policies in Information Security | Acceptable Use Policy (AUP) | – Defines proper use of company systems and networks. | Password Management Policy | – Sets rules for creating and maintaining secure passwords. | Data Backup and Recovery Policy | – Outlines procedures for data protection and restoration. | Incident Response Policy | – Establishes steps to follow during cybersecurity incidents. | Bring Your Own Device (BYOD) Policy | – Regulates the use of personal devices for work. | Access Control Policy | – Determines user roles and permissions for system access.

Slide 10: General Data Protection Regulation (GDPR) | – Protects personal data of EU citizens. | Health | Insurance Portability and Accountability Act (HIPAA) | – Ensures security of healthcare information. | Computer Fraud and Abuse Act (CFAA) | – Criminalizes unauthorized computer access. | Cybersecurity Information Sharing Act (CISA) | – Encourages data sharing for cybersecurity. | Digital Millennium Copyright Act (DMCA) | – Protects digital content and intellectual property. | California Consumer Privacy Act (CCPA) | – Regulates data privacy rights for California residents. | Laws in Information Security

Slide 11: Differences Between Policy and Law | Criteria | Policy | Law | Nature | Internal organizational rule | Government-enforced legal rule | Objective | Guides employees' actions and decisions | Maintains societal order and justice | Authority | Created by an organization | Created by legislative bodies | Applicability | Limited to a specific organization | Applies to all individuals and entities | Modification | Can be updated internally as needed | Requires a legal process to change | Enforcement | Managed by HR or internal teams | Enforced by courts and legal agencies | Consequences | Can result in warnings or termination | Can lead to fines, penalties, or jail | Example | Company cybersecurity policy | Data privacy law (e.g., GDPR, HIPAA)

Slide 12: Law and Ethics in Information Security | Laws: | Rules adopted and enforced by governments to codify expected behavior in modern society | Ethics: | Relatively fixed moral attitudes or customs of a societal group | (based on cultural mores) | The key difference between law and ethics is that law carries the sanction of a governing authority and ethics do not

Slide 13: Law and Ethics in Information Security | Law and ethics guide cybersecurity practices. | Laws ensure data protection and privacy. | Ethics focus on responsible security decisions. | Cyber laws prevent hacking and misuse. | Privacy rights balance security and access. | Compliance ensures trust and legal safety.

Slide 14: Data | Protection | – Ensures confidentiality and integrity of information. | Privacy Regulations | – Safeguards personal and sensitive user data. | Cybercrime Prevention | – Criminalizes hacking, fraud, and cyber threats. | Compliance Standards | – Organizations must follow legal frameworks. | Intellectual Property Protection | – Secures software, patents, and copyrights. | Accountability | – Holds individuals and organizations responsible. | Features of Law and Ethics in Information Security

Slide 15: Types of Laws in Information Security | Data Protection Laws | – Regulate data collection, storage, and processing. | (e.g., GDPR, CCPA, HIPAA) | Cybercrime Laws | – Prevent hacking, fraud, and identity theft. | (e.g., CFAA, UK Computer Misuse Act) | Intellectual Property Laws | – Protect software, trademarks, and copyrights. | (e.g., Copyright Act, Patent Laws) | Compliance and Regulatory Laws | – Ensure businesses follow security standards. | (e.g., ISO 27001, PCI DSS, SOX) | Surveillance and Privacy Laws | – Balance national security with individual privacy. | (e.g., FISA, ECPA, PRISM regulations) | Contract and Liability Laws | – Govern security obligations in business agreements. | (e.g., NDA, SLAs in cybersecurity contracts)

Slide 16: Types of Ethics in Information | Security | Privacy Ethics | – Ensures responsible handling of user data. | Hacking Ethics | – Differentiates ethical hacking (white-hat) from malicious hacking (black-hat). | AI and Security Ethics | – Manages bias, transparency, and accountability in security AI. | Corporate Ethics | – Defines security responsibilities of organizations. | Whistleblowing Ethics | – Protects individuals reporting security violations. | Cybersecurity Professional Ethics | – Guides IT professionals on ethical security practices. | (e.g., ISC² Code of Ethics)

Slide 17: Examples of Law and Ethics in Information Security | Facebook-Cambridge | Analytica | Scandal (Privacy Ethics & Data Protection Laws Violation) | WannaCry | Ransomware Attack (Cybercrime & Compliance Failure) | Apple vs. FBI Encryption Case (Privacy vs. Surveillance Ethics) | Google AI Bias in Hiring (AI and Security Ethics Concern) | Edward Snowden NSA Leaks (Whistleblowing & Government Surveillance Ethics) | Sony Pictures Hack (Intellectual Property & Cybercrime Violation)

Slide 18: Advantages and Disadvantages of Law and Ethics in Information Security | Advantages: | Data Protection | – Ensures confidentiality and prevents unauthorized access. | Privacy Assurance | – Strengthens user trust in organizations. | Cybercrime Prevention | – Reduces risks of hacking and fraud. | Legal Compliance | – Avoids lawsuits and regulatory penalties. | Corporate Accountability | – Holds businesses responsible for security breaches. | Ethical Decision-Making | – Encourages responsible cybersecurity practices.

Slide 19: Disadvantages: | Compliance Costs | – Implementing security laws can be expensive. | Restrictive Policies | – May limit innovation and data access. | Legal Complexity | – Varying global regulations make compliance challenging. | Privacy vs. Security Conflict | – Government surveillance may compromise individual rights. | Slow Legal Updates | – Laws may not keep pace with new cyber threats. | Enforcement Challenges | – Difficult to track and penalize cybercriminals globally

CSE332 Lecture 21.pptx (16 slides)

Slide 1: CSE332 | Unit | 6 | :- | Ethical and Professional issues in Information Security

Slide 2: Need for Cyber Law in India | Cyber law refers to the legal framework that governs | cybercrime, digital transactions, data protection, and online activities. | It defines the rights, responsibilities, and punishments related to crimes committed in cyberspace.

Slide 3: Objectives of Cyber Law: | ☒ Prevent cybercrimes and punish offenders. | ☒ Protect individuals, businesses, and government entities from cyber threats. | ☒ Establish guidelines for e-commerce, digital payments, and IT security. | ☒ Safeguard data privacy and prevent unauthorized access. | ☒ Regulate electronic evidence in legal matters.

Slide 4: Key Cyber Laws in India | Information Technology (IT) Act, 2000 | ♦ India's primary cyber law that governs cybercrimes, digital transactions, and electronic evidence. | ♦ Covers: Hacking, identity theft, phishing, cyberstalking, and data protection. | ♦ Punishments: Up to 10 years of imprisonment and fines for serious cyber offenses.

Slide 5: Indian Penal Code (IPC) – Sections Related to Cybercrime | ♦ | Section 419 & 420: | Punishment for cheating and online fraud. | ♦ | Section 500: | Punishment for cyber defamation. | ♦ | Section 354D: | Covers cyberstalking and online harassment.

Slide 6: Digital Personal Data Protection (DPDP) Act, 2023 | ♦ Protects | personal data privacy | and regulates how companies collect and use user information. | ♦ Ensures strict penalties for | data breaches and misuse of private data.

Slide 7: National Cyber Security Policy, 2013 | ♦ Strengthens India's cybersecurity infrastructure. | ♦ Focuses on protecting | critical sectors like banking, defense, and telecom.

Slide 8: Challenges in Implementing Cyber Laws | △ | Lack of Awareness | – Many people are unaware of cyber laws and their rights. | △ | Jurisdiction Issues | – Cybercrimes often involve multiple countries, making enforcement difficult. | △ | Slow Legal Proceedings | – Cybercrime cases take time to resolve due to technical complexities. | △ | Cybercriminals Using Advanced Technologies | – AI-powered malware and the | dark web | make it harder to track criminals.

Slide 9: Ethical and Professional issues in Information Security | Cyber Crime ,Cyber-crime on the rise, Need for cyber law in India

Slide 10: Cyber crime | Cybercrime refers to any illegal activity that involves computers, networks, or digital devices. | These crimes are committed using technology to steal, manipulate, or destroy data, systems, or personal information. | Cybercrimes can target individuals, businesses, or even governments, leading to financial losses, security breaches, and privacy violations.

Slide 11: Types of Cyber Crimes | Cyber Fraud - | Any fraudulent activity conducted online to deceive individuals or businesses for financial gain. | Hacking & Data Breaches | - Gaining unauthorized access to computer systems or networks to steal, modify, or delete data. | Cyberbullying & Harassment | - The use of digital platforms to harass, threaten, or abuse someone. This includes sending offensive messages, spreading false rumors, or sharing private photos without consent.

Slide 12: Intellectual Property Theft - | The unauthorized use, reproduction, or distribution of copyrighted material, trademarks, or patented content. | Cyber Terrorism | - The use of cyber attacks to disrupt critical infrastructures, government systems, or spread fear among the public.

Slide 13: Cyber-Crime on the Rise | Cybercrime has been increasing at an alarming rate worldwide. | With the rapid growth of digital transactions, cloud computing, and social media. | Cybercriminals have found new ways to exploit individuals, businesses, and governments. | The rise in cybercrime can be attributed to greater internet penetration, lack of cybersecurity awareness, and evolving hacking techniques.

Slide 14: Statistics and Reports on Rising Cybercrime | Cybercrime is expected to cost the world \$10.5 trillion annually by 2025 (Cybersecurity Ventures report). | A cyberattack occurs every 39 seconds worldwide. | In 2023, more than 800 million ransomware attacks were reported globally. | Over | 50,000 cybercrime cases | were registered in India in 2023 (NCRB Data).

Slide 15: Major Reasons Behind the Rise in Cybercrime | Increased Digital Dependence | Lack of Cybersecurity Awareness | Advanced Hacking Techniques | Rise of Mobile & IoT Devices | (| 43% of cyberattacks | are now directed at mobile devices. |) | Financial Motivation

Slide 16:

CSE332 Lecture 23.pptx (11 slides)

Slide 1: CSE332 | INDUSTRY ETHICS AND LEGAL ISSUES | Hands on tools for | cyber security), | case | studies on | cyber crimes)

Slide 2: Emerging Trends in Cybersecurity Tools | Stay ahead with emerging tools. AI-powered threat detection offers predictive analytics. | Secure | Access Service Edge (SASE) enhances network security. | Password less | authentication uses biometrics. Cloud-focused tools protect multi-cloud environments.

Slide 3: Cybersecurity is continuous, not a one-time fix. Combine tools, processes, and trained people. | Adopt | new tools, upgrade existing systems, and educate your team to stay ahead. | Conclusion: Staying Ahead in Cybersecurity

Slide 4: The Growing Importance of Cybersecurity | Cybersecurity is more critical than ever. | By 2025, global damage from cybercrime is expected to reach | \$10.5 trillion | . | Ransomware attacks | rose by | 13% in 2023 | . | An estimated | 68% of business leaders | feel their cybersecurity risks are increasing. | Learn how to stay secure.

Slide 5: Essential Cybersecurity Tools for Protection | Use these essential tools to enhance your cybersecurity: | Firewalls | provide a first line of defense. | VPNs | protect data in transit by encrypting connections. | Anti-virus software | prevents known threats. | Intrusion Detection Systems (IDS) | provide real-time threat monitoring.

Slide 6: Cybersecurity Tools | 1. Network Security Tools: | - | Wireshark, | Nmap | , Snort, | Tcpdump | 2. Endpoint Security Tools: | - | CrowdStrike | Falcon, Symantec, Bitdefender | GravityZone | 3. Penetration Testing Tools: | - | Metasploit | , Burp Suite, Nessus, Kali Linux | 4. Forensics & Investigation Tools: | - | Autopsy, FTK Imager, Volatility | 5. SIEM Tools: | - | Splunk | , IBM | QRadar | , ELK Stack

Slide 7: Employ advanced tools for robust defense | : | Endpoint Detection and Response (EDR) | identifies and responds to endpoint threats. | Vulnerability scanners | detect system weak points. | Encryption tools | secure sensitive data. | Security Information and Event Management (SIEM) | offers holistic monitoring. | Advanced Tools for Proactive Defense

Slide 8: Case Study 1: Colonial Pipeline Ransomware Attack (2021) | The | Colonial Pipeline attack | disrupted the U.S. fuel supply. | Hackers received a | \$4.4 million Bitcoin payment | . | This case highlights the

importance of: | Multi-Factor Authentication (MFA) | Network segmentation | Take steps now to ensure you are secure.

Slide 9: Case Study 2: Email Phishing Against a Fortune 500 Company | A | phishing attack | led to a | \$100 million loss | in wire fraud. | The attack vector involved | deceptive emails impersonating executives | . | To prevent email scams, implement: | Staff training | SPF, DKIM, and DMARC | email authentication protocols

Slide 10: Case Study 3: Data Breach at Equifax (2017) | The | Equifax data breach | compromised personal data of | 147 million people | . | Cause: An | unpatched vulnerability | in | Apache Struts | software. | This case illustrates the critical need for: | Timely software updates | Effective patch management

Slide 11: Best Practices for Cybersecurity Resilience | Regular software updates and patch management | Implement Multi-Factor Authentication (MFA) | Conduct employee cybersecurity awareness training | Use strong, unique passwords and a password manager | Backup critical data regularly and securely | Utilize firewalls, antivirus, and endpoint protection | Perform regular vulnerability assessments and penetration testing | Enforce least privilege access controls | Deploy network segmentation and monitoring tools | Establish a robust incident response and recovery plan